MILITARY SPECIFICATION

HEADSET-MICROPHONE, MONAURAL, LIGHTWEIGHT, INTRA-AURAL GENERAL SPECIFICATION FOR

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers intra-aural headsets intended for use in low ambient noise environments below 85 dBA SPL (sound pressure level). These headset assemblies shall use magnetic earphones that transform electrical energy into acoustical energy and shall use electret microphones to transform voice acoustical energy into electrical energy. The headset shall be a completed assembly including an earphone capsule with internal earphone element, externally mounted earphone acoustic tube, internal microphone element with externally mounted acoustic voice tube, microphone, earphone amplifiers with earphone volume control, cord, connector, and amplifier configurations to meet the needs of various military activities. Power to the connector tips is required for headset operation. The headsets are designed to be worn over the ear or attached to the user's eyeglass frames.

1.2 Part number. The part number shall be in accordance with the applicable specification sheet (see 6.7).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks.. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Strapping, Steel, and Seals.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Electronic Support Division AFLC, 2750 ABW/ES, Gentile Air Force Station, Dayton, OH 45444 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5965

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-F-14256	-	Flux, Soldering, Liquid (Rosin Base).
MIL-H-83534/1(USAF)	-	Headset-Microphone, Monaural, Lightweight, Intra-aural, Amplified Receiver.
MIL-H-83534/2(USAF)	-	Headset-Microphone, Monaural, Lightweight, Intra-aural, Unamplified Receiver.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.
MIL-STD-2073-1	-	DOD Materiel Procedures for Development and Application of Packaging Requirements.
MIL-STD-45662	-	Calibration Systems Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. The headset or headset-microphone assemblies furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for award of contract (see 4.6 and 6.5).

3.3 Materials.

3.3.1 Flux soldering, liquid (rosin base). The solder flux shall conform to type R of MIL-F-14256.

3.4 Design and construction.

3.4.1 Headsets. The assemblies specified herein shall not cause dermatitis by reason of selection, processing, or fabrication of components. The headset shall be designed to be worn by all military personnel, including personnel wearing eyeglasses. The headset assembly shall be specified (see 3.1). A typical configuration is as shown on figure 1.

3.4.1.1 Weight. The maximum weight of the entire headset assembly shall be 12 ounces.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.

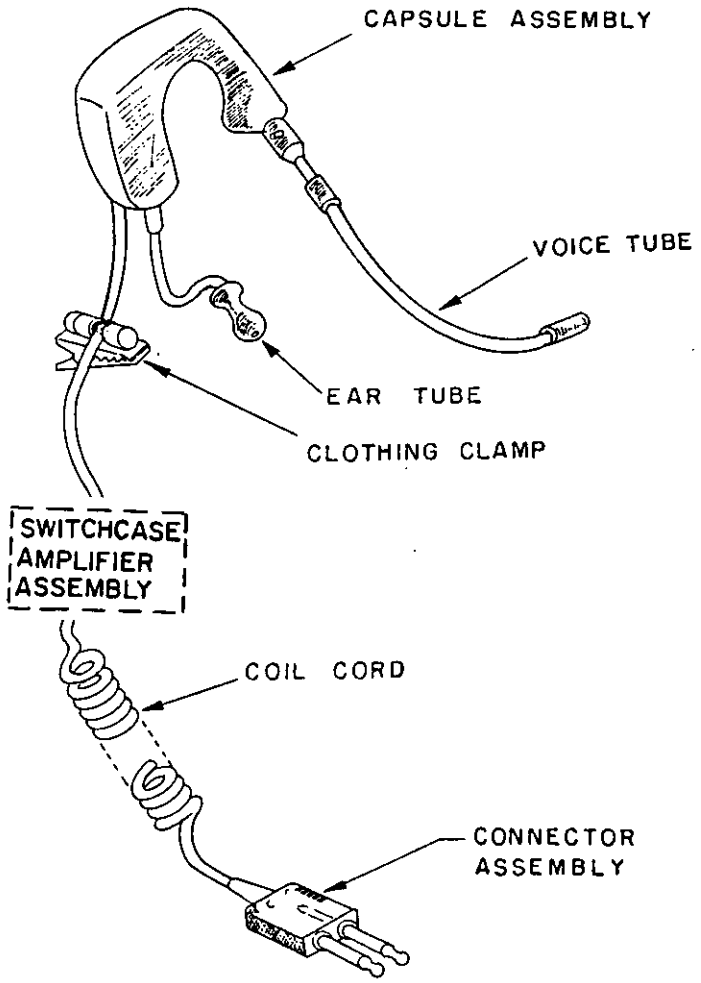


FIGURE 1. Typical headset assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

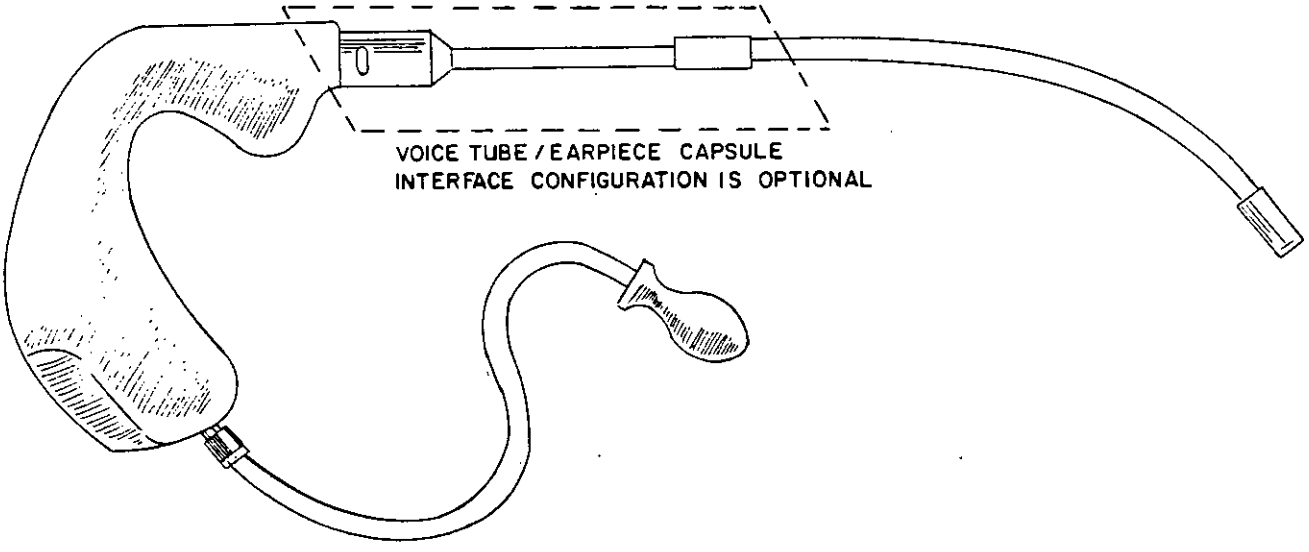


FIGURE 2. Headset-microphone capsule assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.

3.4.1.1.1 Headset weight. The maximum weight of the headset portion not including the cable, plug, and amplifier assemblies shall be .5 ounce.

3.4.1.2 Headset color. Unless otherwise specified (see 3.1), the color of all exterior nonmetalic surfaces shall be black Cycolac ABS, except the voice tube and ear tube which are transparent.

3.4.1.3 Metals. The metals used shall be of the proper alloy and hardness necessary to provide the required strength and rigidity with the maximum strength to weight ratio. Metals shall be of a corrosion-resistant type.

3.4.2 Headset-microphone capsule assembly. The earphone with microphone capsule assembly shall be as shown on figure 2.

3.4.2.1 Earphone and microphone element location. Both the earphone element and the microphone element shall be located in the earphone capsule. Voice signals shall be transmitted to the microphone element through fully adjustable tubulation. Received signals shall be sent acoustically from the earphone element to the user through flexible tubulation terminated with an ear tip which couples to the user's ear canal (see figure 3 for typical configuration when in place on user's head). Ear tip and tubulation shall be permanently affixed to each other and shall not separate under normal use.

3.4.2.2 Voice tube adjustability. The voice tube shall be adjustable to allow placement of the voice tube or microphone element for optimum voice transmission (see figures 3 and 4).

3.4.2.3 Ear tube and ear tip configurations. The ear tubes and ear tips shall be shown on figure 5.

3.4.2.4 Earphone capsule materials. The earphone capsule (see figure 2, item 5) shall be fabricated from Acrylonitrile-Butadiene-Styrene (ABS) plastic or equivalent.

3.4.2.5 Microphone element. Microphone element shall be an electret condenser microphone.

3.4.2.6 Earphone element. Earphone element shall be a magnetic transducer.

3.4.2.6.1 Earphone element mounting. The earphone element shall be mounted to provide for shock protection.

3.4.2.7 Earphone capsule shape. The earphone capsule shall be contoured as shown on figure 6.

3.4.3 Cable assemblies. Unless otherwise specified (see 3.1), there shall be two cables provided with each headset assembly as shown on figure 7. The capsule cable assembly shall connect the earphone assembly with the amplifier assembly, and incorporate an adjustable clothing clamp to secure the cable to the users clothing. The connector cable shall connect the amplifier assembly with the connector assembly. The connector cable shall be of the retractile type, length as specified (see 3.1).

3.4.4 Connector assembly. Unless otherwise specified (see 3.1), the connector assembly shall be as shown on figure 8. The connector shall be a Western Electric Company 425 or equivalent configuration.

3.4.5 Amplifier assembly. Unless otherwise specified (see 3.1), the amplifier assembly shall be configured approximately as shown in figure 9. The amplifier assembly shall contain the microphone and earphone amplifiers, an earphone volume control switch (three position or continuously variable), microphone push-to-talk switch, push-to-talk locking switch, a belt clip, and wrist loop as follows.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.

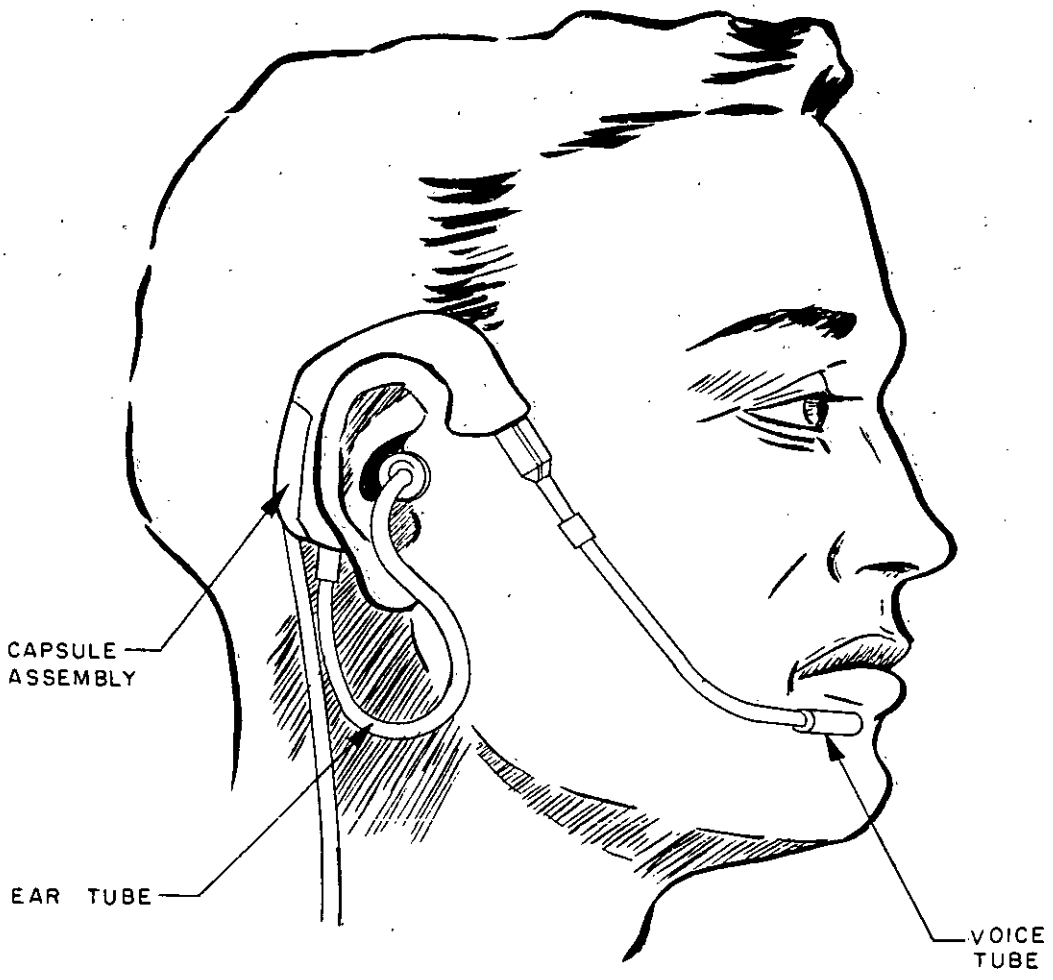
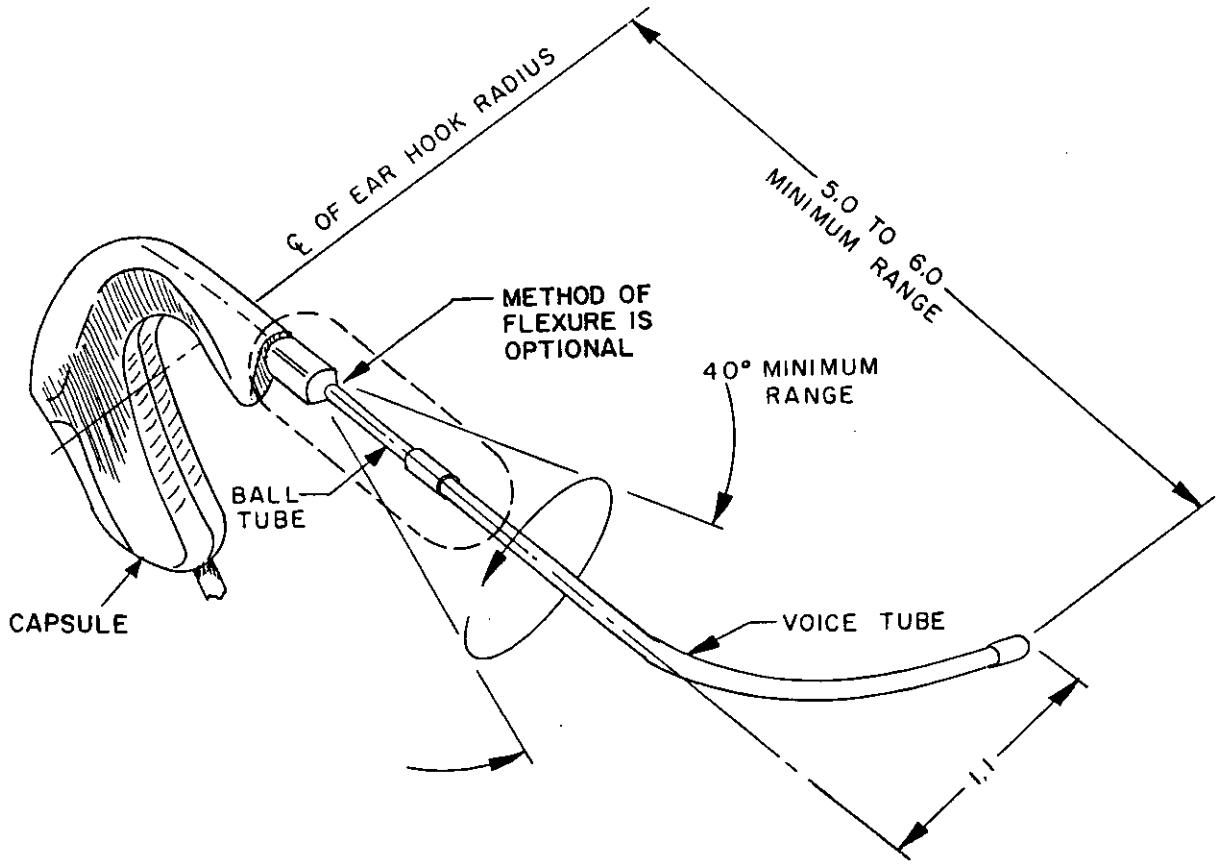


FIGURE 3. Headset-microphone when in place.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.



NOTES:
1. The voice tube shall be rotatable 360° about the ball tube axis.
2. The ball tube shall be adjustable anywhere within a 40° conical volume as shown.
3. The voice tube shall be adjustable such that the distance between the tip and the center of the ear hook radius be set within a minimum range of 5.0 to 6.0 inches as shown.

FIGURE 4. Voice tube adjustability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

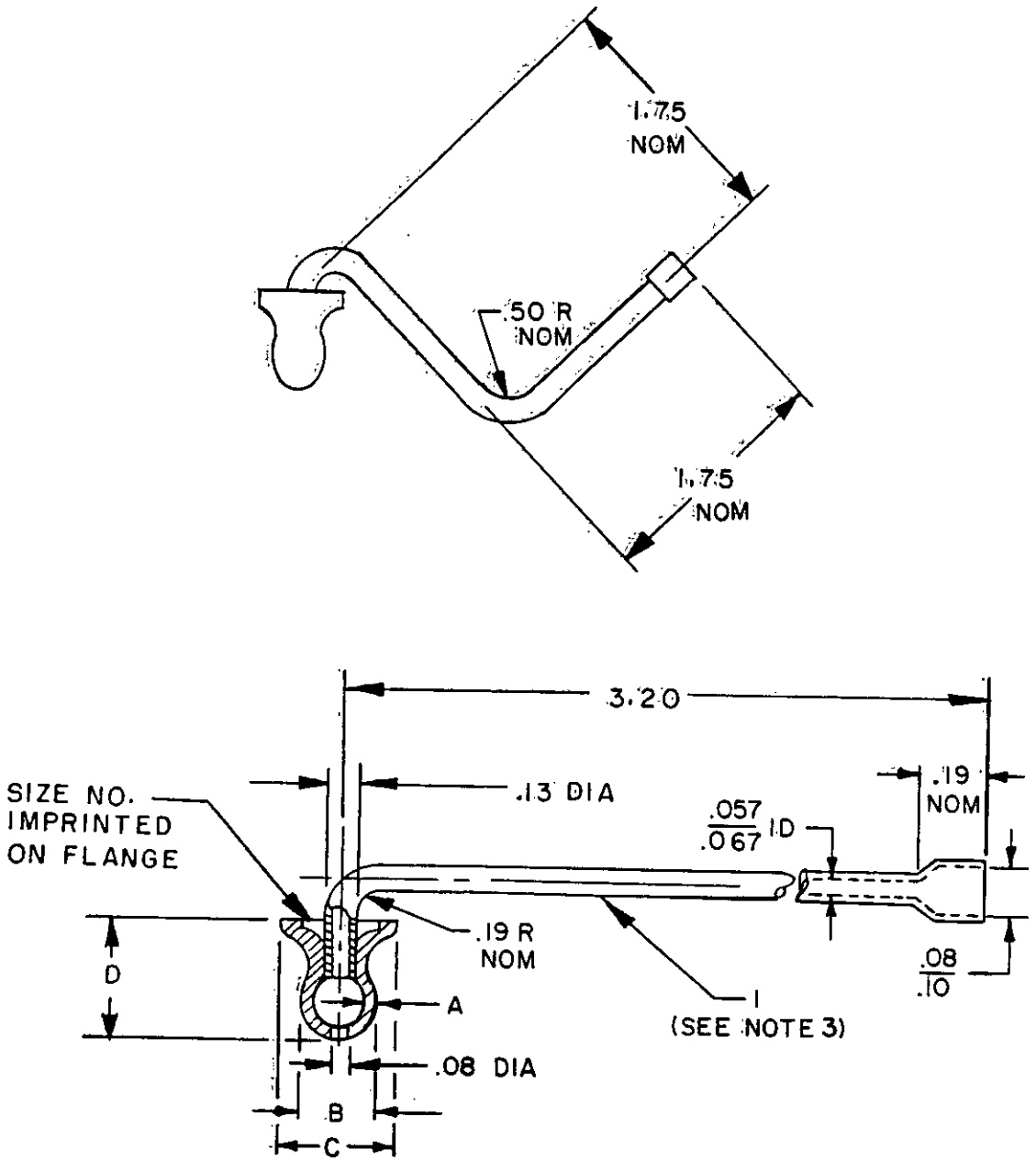


FIGURE 5. Ear tube and ear tip configurations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

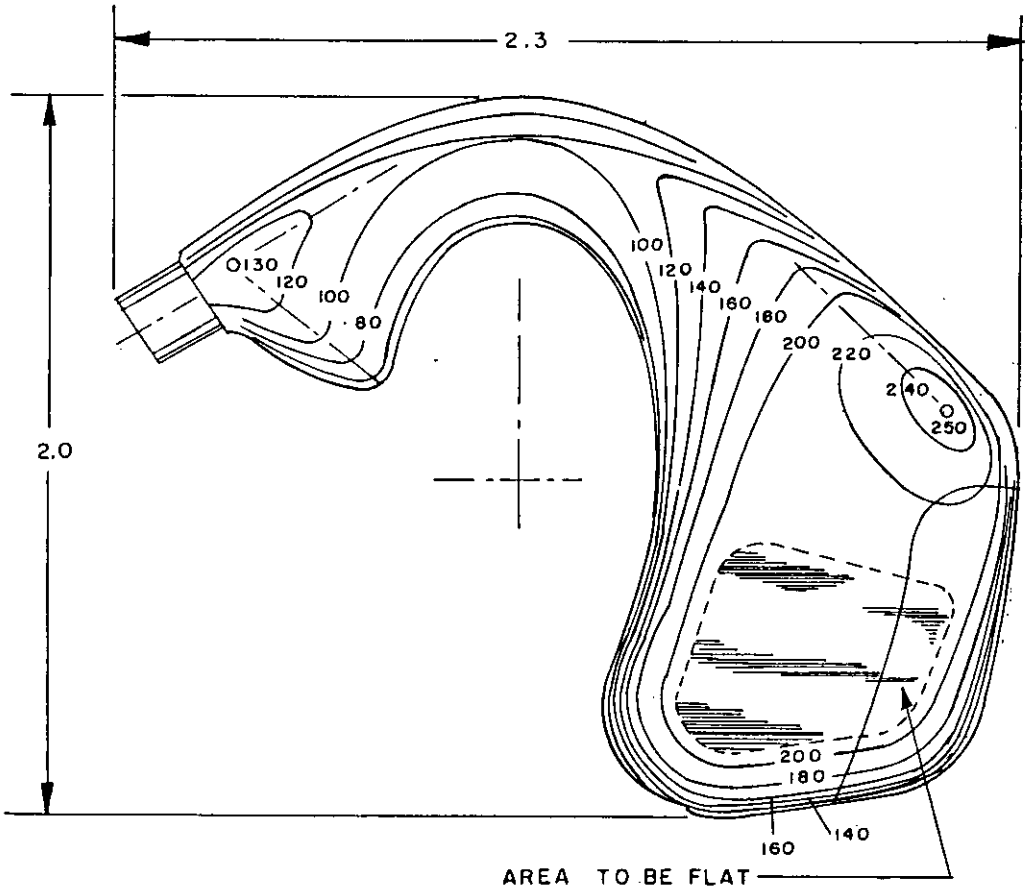
Size number	Dimensions			
	Nominal A	Dia B ±.02	Dia C ±.04	D ±.04
1	.025	.23	.34	.42
2	.030	.26	.38	.52
3	.035	.29	.42	.57
4	.040	.32	.46	.62
5	.045	.36	.50	.67
6	.060	.38	.54	.77

Inches	mm	Inches	mm
.025	0.64	.32	8.1
.03	0.8	.34	8.6
.035	0.89	.35	8.9
.040	1.02	.37	9.4
.045	1.14	.38	9.7
.05	1.3	.42	10.7
.057	1.45	.46	11.7
.06	1.5	.50	12.7
.067	1.70	.54	13.7
.08	2.0	.62	15.7
.09	2.3	.65	16.5
.10	2.5	.77	19.6
.13	3.3	1.12	28.4
.19	4.8	1.75	44.5
.22	5.6	1.875	47.63
.23	5.8	1.96	49.8
.24	6.1	2.12	53.8
.25	6.4	2.75	69.9
.26	6.6	2.85	72.4
.27	6.9	3.0	76.2
.29	7.4	3.20	81.3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Each tip shall be firmly adhered to its own tube.
4. Ear tip size shall be molded on the underside of ear tip flange. Size 1 is smallest.
5. Earpiece tube: Clear TYGON, or equivalent, 0.62 ID tubing ordorless, tasteless, and non toxic.
6. Earpiece tip: Pink PLASTISOL, or equivalent, nonallergenic.

FIGURE 5. Ear tube and ear tip configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

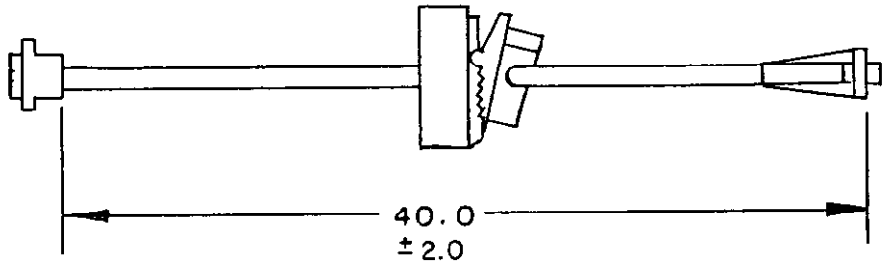


Inches	mm
1.0	25.
1.875	47.63
5.25	133.4

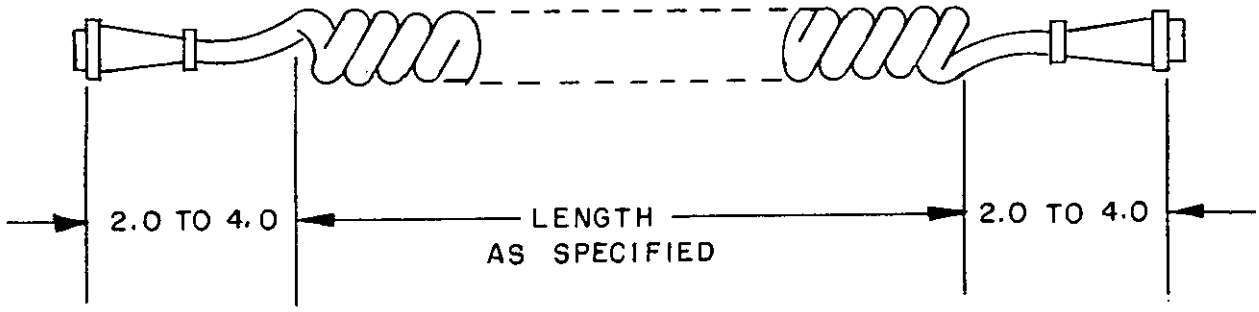
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Numbers on contour lines are elevation in mils from center plane.
4. Earphone capsule is symmetrical with respect to center plane.
5. Configuration is optional. These dimensions provide a comfortable fit.

FIGURE 6. Typical capsule contour.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.



CAPSULE CABLE



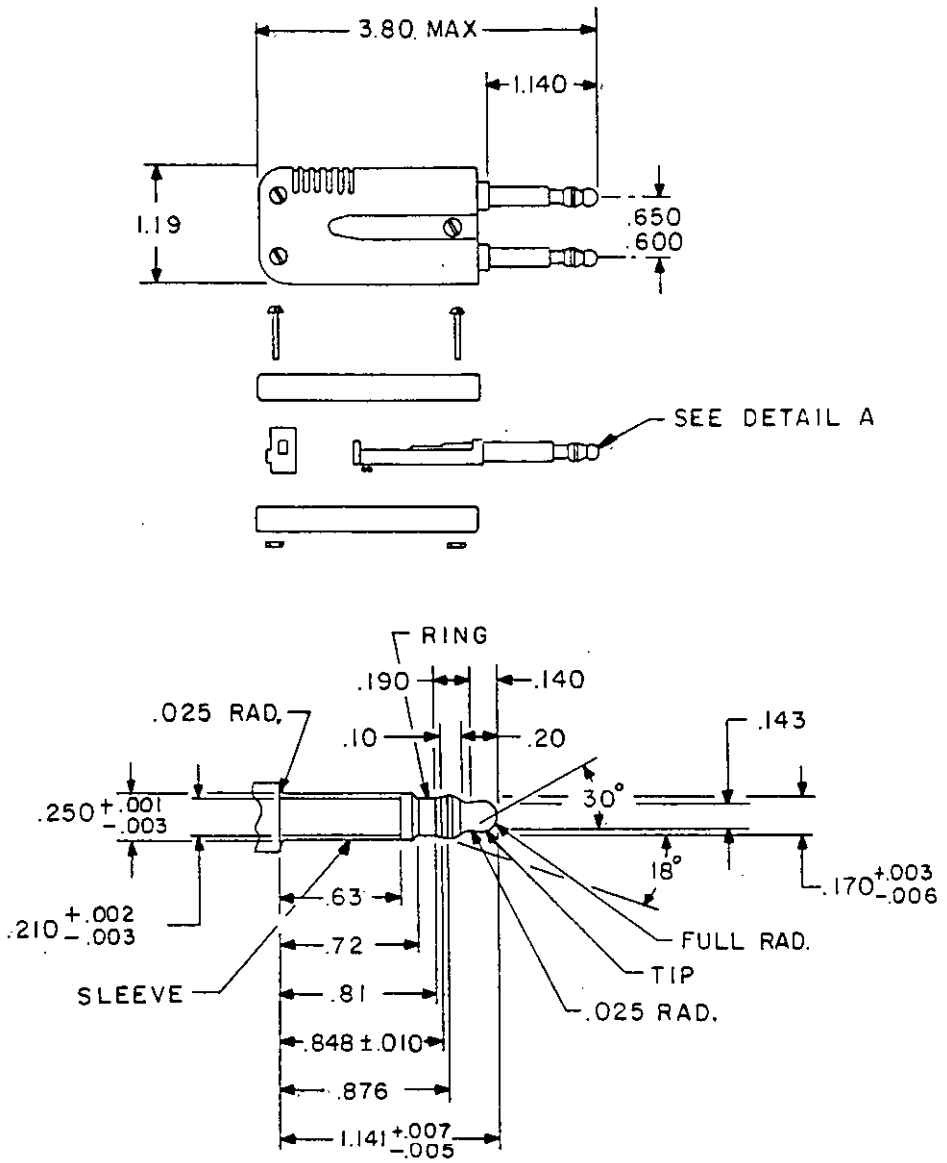
Inches	mm
2.0	50.8
4.0	101.6
40.0	1016.

CONNECTOR CABLE

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only.

FIGURE 7. Cable assemblies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

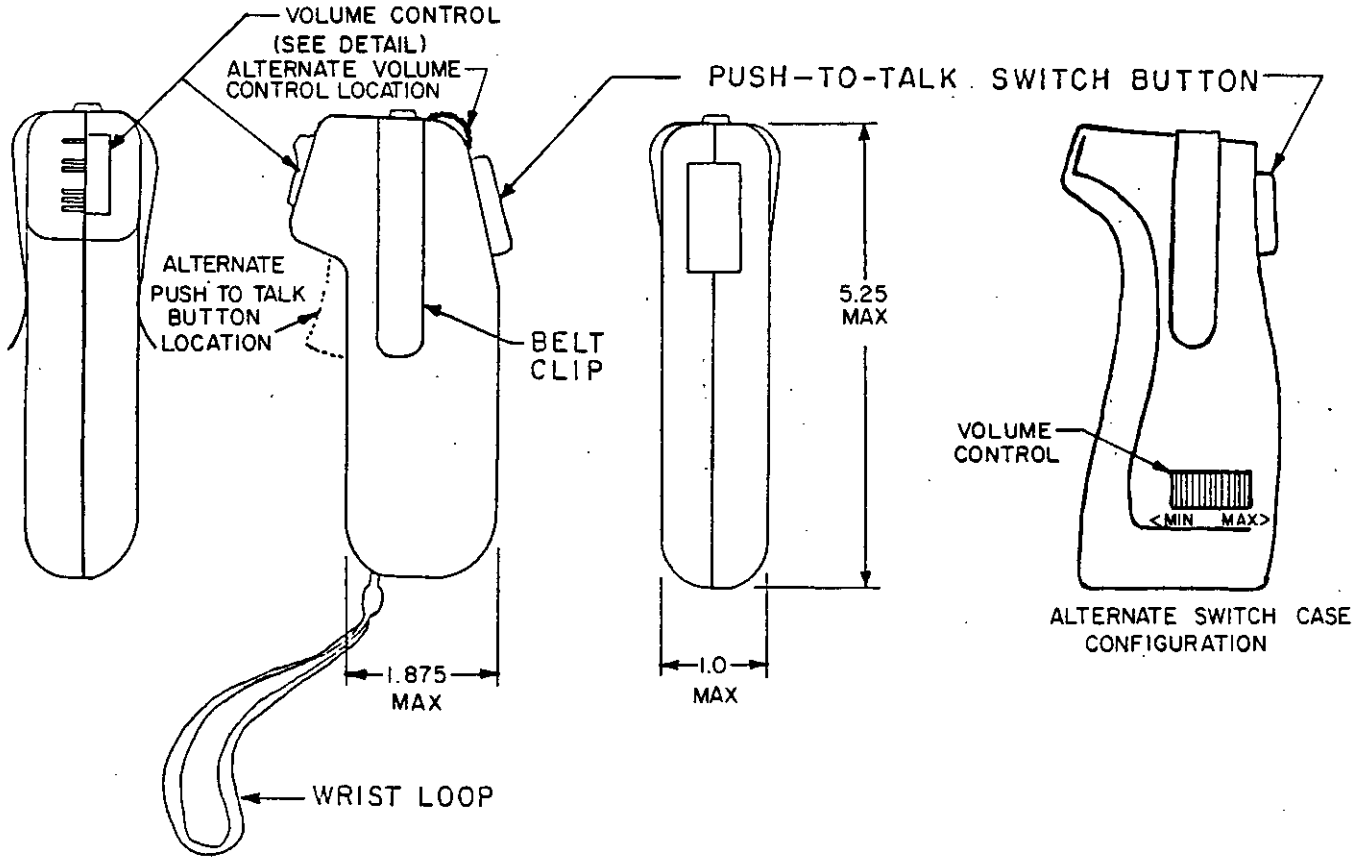


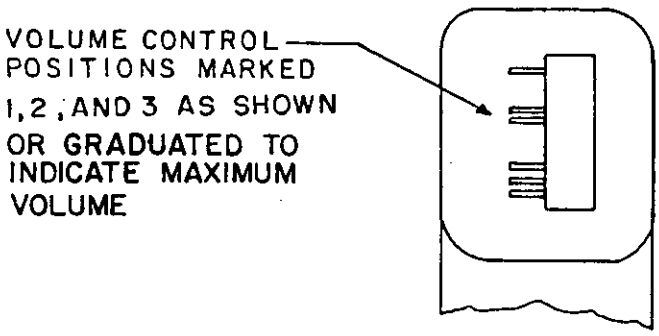
Inches	mm
.001	0.03
.002	0.05
.003	0.08
.005	0.13
.006	0.15
.007	0.18
.010	0.25
.025	0.64
.10	2.5
.140	3.56
.143	3.63
.170	4.32
.190	4.83
.20	5.1
.210	5.33
.250	6.35
.600	15.24
.63	16.0
.650	16.51
.72	18.29
.81	20.6
.848	21.54
.876	22.25
1.140	28.96
1.141	28.98
1.19	30.2
3.80	96.52

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 8. Connector assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.





Inches	mm
1.0	25.4
1.875	47.63
5.25	133.4

VOLUME CONTROL DETAIL

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Locations of volume control and push-to-talk buttons are left to the manufacturer's option.

FIGURE 9. Switch case amplifier assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

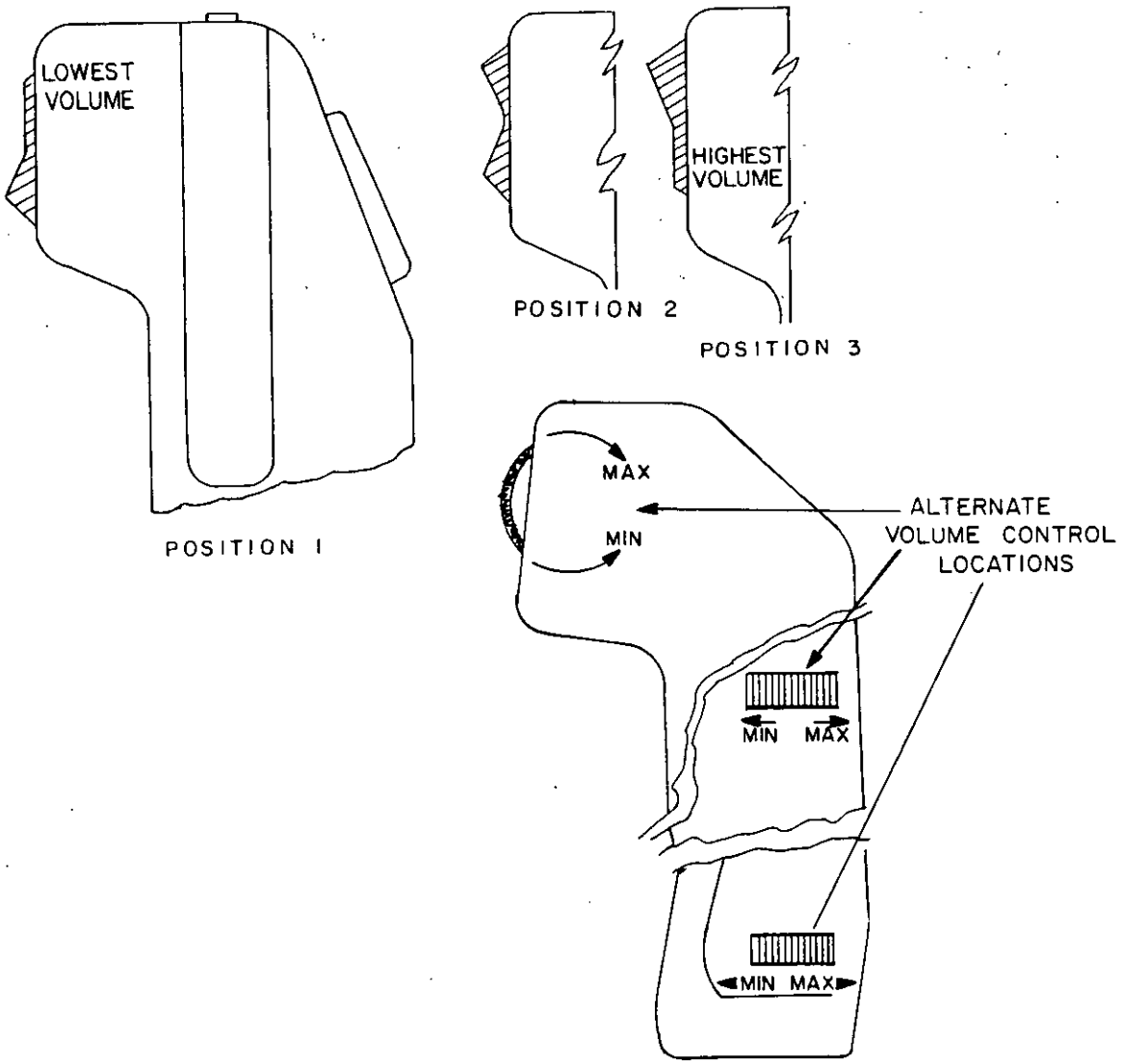


FIGURE 10. Volume control operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

3.4.5.1 Receiver circuit. The receiver amplifier circuit shall include receive limiting, a receiver amplifier with a volume control, and receiver compression circuit.

3.4.5.2 Transmit circuit. The transmit circuit shall include an amplifier, voice switching and polarity circuitry. The transmit circuit shall include a push-to-talk switch as specified in 3.4.5.4.

3.4.5.3 Volume control switch operation. The volume control shall be either a three-position rocker switch or a continuously variable thumb wheel rotary potentiometer.

3.4.5.3.1 Life. The volume control shall operate a minimum of 100,000 cycles without failure in any of the three switch positions for rocker type volume controls and 25,000 cycles full range for potentiometer type volume controls.

3.4.5.3.2 Volume control switch operation. If the switch is a three-position rocker type, it shall operate as shown on figure 10.

3.4.5.4 Push-to-talk switch. The push-to-talk switch shall meet the following requirements:

3.4.5.4.1 Life. The switch shall operate a minimum of 100,000 cycles without failure.

3.4.5.4.2 Actuator force. The force necessary, when exerted on the switch button (see figure 11), to activate the switch (fully depressed) shall be 1 to 2 pounds.

3.4.5.4.3 Contact and locking sequence. The switch shall provide operation (see figure 11) as follows:

a. Position 1: Switch not depressed, for monitoring telephone or radio.

b. Position 2: Switch depressed, for push-to-talk transmit operation.

c. Position 3: Switch depressed and locked, for locked-on transmit operation. The locking/non-locking push-to-talk option shall not be user selectable, but shall be accessed only at the service/maintenance level.

3.4.5.4.4 Switch contact gap. The gap between the switch contacts shall be between .015 and .035 inch when in the off position.

3.4.5.4.5 Switch button. The switch button shall be as shown on figure 12.

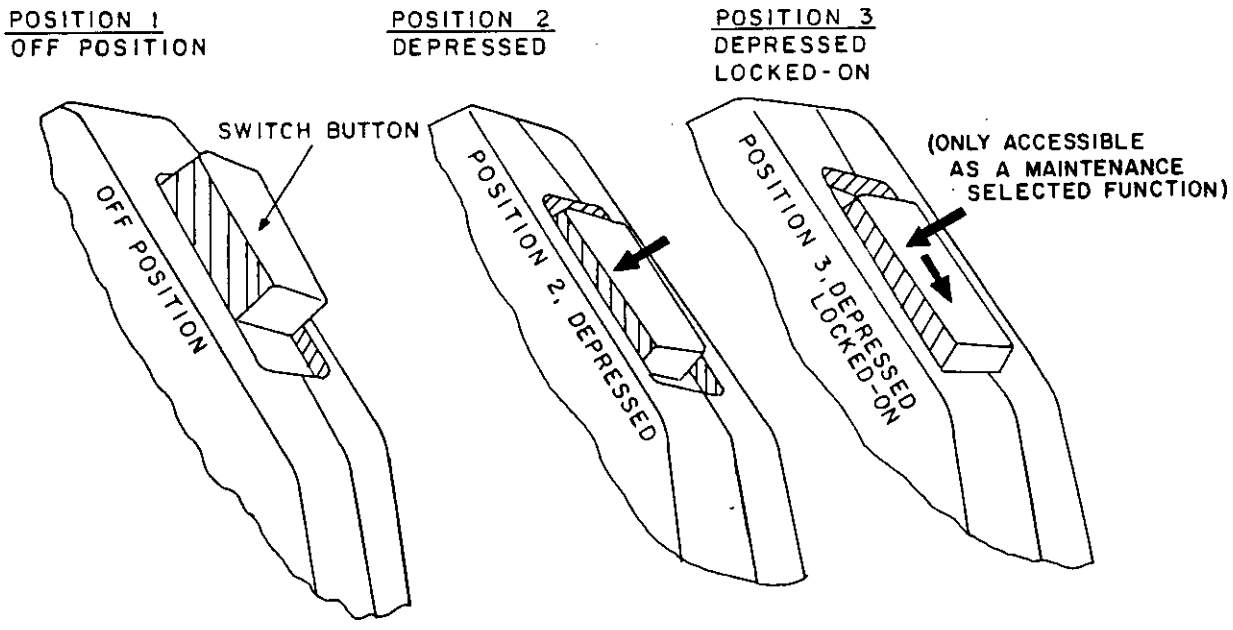
3.4.5.5 Locking switch. The locking switch shall control the locking/non-locking feature of the push-to-talk switch (see 3.4.5.3), and shall not be user accessible. This function shall be activated at the service/maintenance level only. When in the non-locking position, Position 3 of the push-to-talk switch (see figure 11) shall not be achievable.

3.4.5.6 Switch case. The switch case shall be contoured to allow comfort during extended periods of handheld use.

3.4.5.7 Wrist loop. The wrist loop shall be configured approximately as shown on figure 13.

3.4.5.8 Belt clip. The belt clip shall be provided (see figure 14).

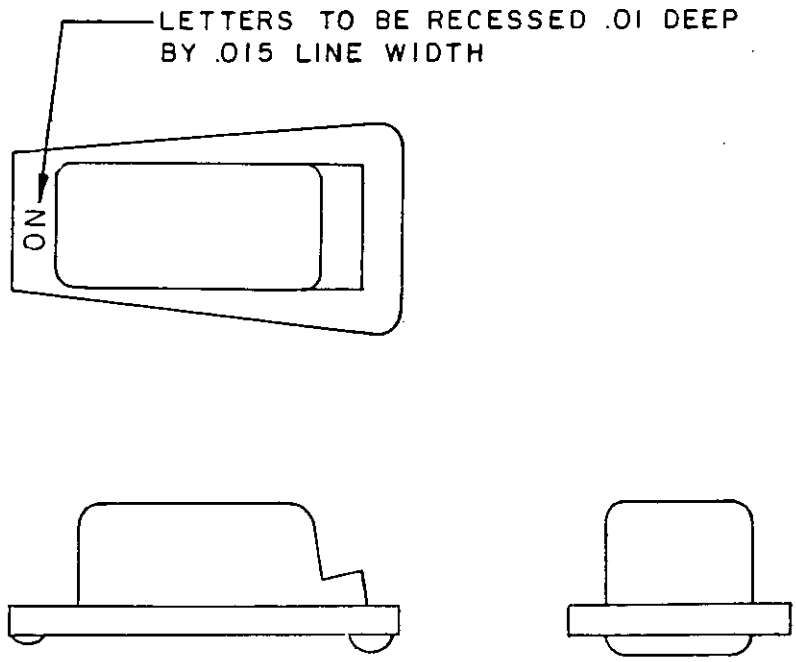
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.



NOTE: To lock-on (position 3) the switch button shall be depressed and moved as indicated by arrows.

FIGURE 11. Push-to-talk switch operation.

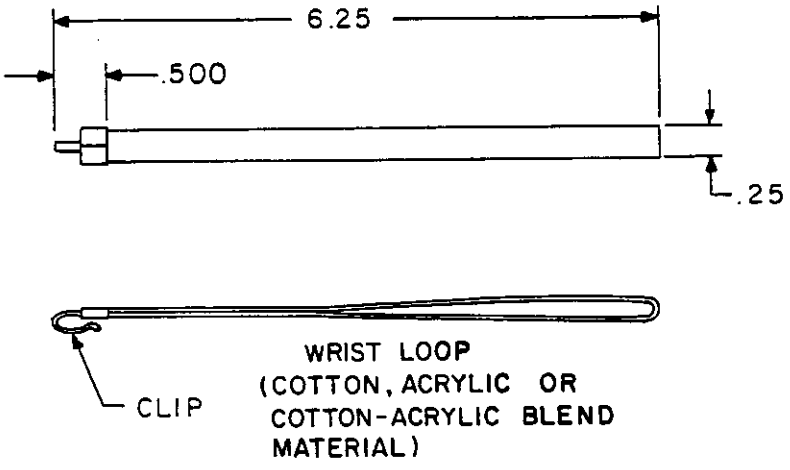
Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.



MATERIAL : PLASTIC
COLOR : WHITE
SURFACE FINISH : SPI NO.2 OR BETTER

FIGURE 12. Switch button.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

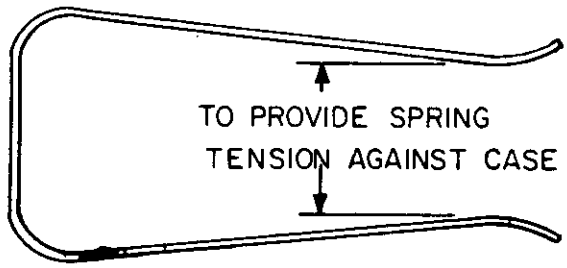


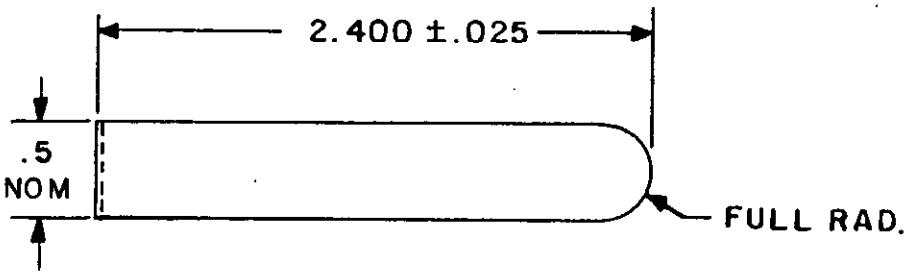
Items:	Inches	mm
1. Clip	.25	6.4
2. Wrist loop	.500	12.7
	6.25	158.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are for general information only.

FIGURE 13. Wrist loop.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.



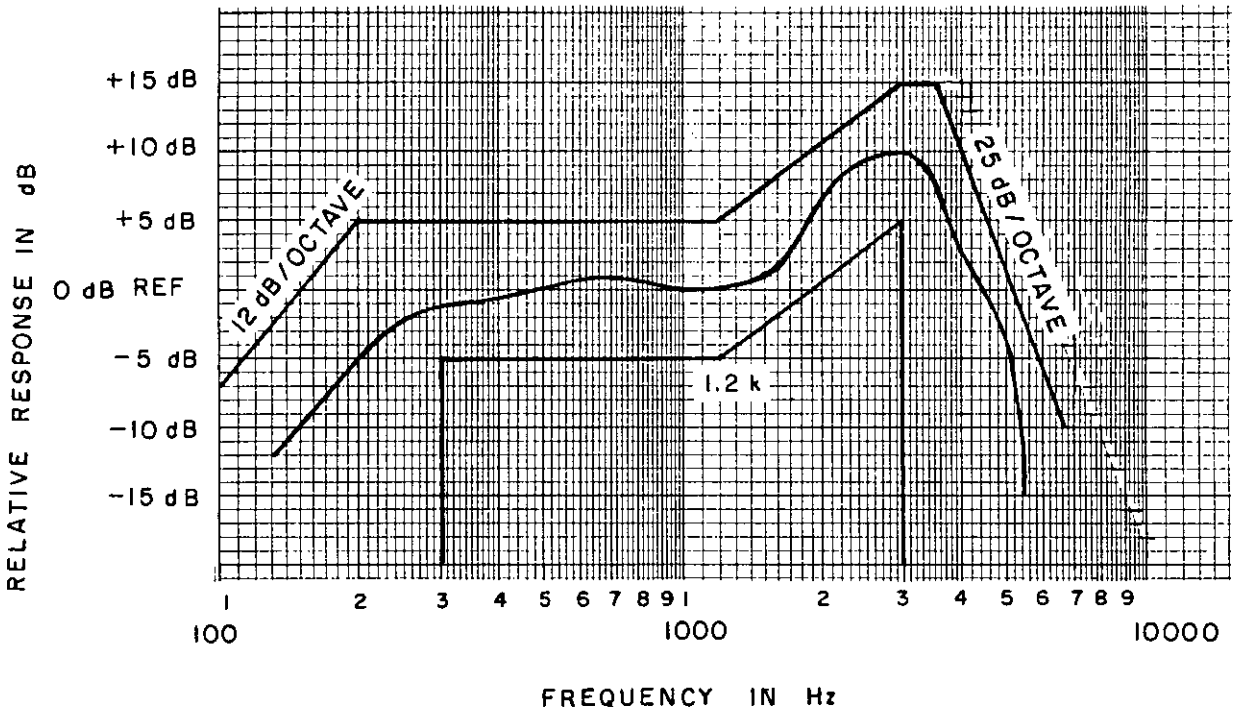


Inches	mm
.25	6.4
.5	13.
2.40	60.96

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Fabricate from 301 stainless steel full hard.
4. Finish: Bright polish.
5. Exterior of part to tbe free of tooling marks. Remove all burrs and sharp edges.

FIGURE 14. Belt clip.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.



NOTE: The response curve shown is typical and may "float" within the envelope limits.

FIGURE 15. Transmit response envelope.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.

3.5 Transmit performance characteristics. Transmit performance characteristics shall be measured under conditions specified in 4.8.2. The transmitter includes the microphone element and microphone amplifier. All transmit specifications shall apply for each polarity of transmitter dc current. The maximum difference in output level from positive to negative current operation shall be 1.5 dB.

3.5.1 Transmit output level. When tested as specified in 4.8.2, the 1,000 Hz output level shall be -22.5 ±4 dBV. When tested with an electrically similar headset connected in parallel, the 100 Hz output level shall be within 6.5 dB of the value measured as a single headset.

3.5.2 Transmit loudness level (TXLL) (reference only). When tested as specified in 4.8.3, the loudness level shall be -22 ±4 dB. When tested with an electrically similar headset connected in parallel, the loudness level shall be within 6.5 dB of the value measured as a single headset.

3.5.3 Frequency response. When tested as specified in 4.8.4, the frequency response shall fall within the envelope of figure 15. The response curve may float within the envelope and touch, but not cross, the limits of the envelope.

3.5.4 Voice switching (see 4.8.5).

3.5.4.1 Switch point. Voice switching shall occur between acoustic drive levels of 72.5 dB SPL and 82.5 dB SPL when tested as specified in 4.8.5.1.

3.5.4.2 Switch depth and time. The transmit level with acoustic drive level below switch point shall be 10 to 17 dB less than it is with the drive level above switch point when tested as specified in 4.8.5.2. The switch-up shall occur within 5 to 15 milliseconds. The switch-down shall occur within 150 to 300 milliseconds.

3.5.5 Warm-up time. When tested as specified in 4.8.6, the transmit output shall reach 71 percent of the steady state value within 50 milliseconds of the time that 90 percent of steady state power is received at the tips.

3.5.6 Distortion. Total harmonic distortion shall be 5 percent or less when tested as specified in 4.8.7.

3.5.7 DC resistance. The dc resistance in ohms shall be 50 ohms or greater in test condition 1 and 40 ohms or greater in test condition 4 when tested as specified in 4.8.8.

3.5.8 Output impedance. When tested as specified in 4.8.9, the output impedance shall be 50 ohms ±20 ohms at all current levels.

3.5.9 Transmit noise. The average noise level shall be 10 dBrnC or less when tested as specified in 4.8.10.

3.6 Receiver performance characteristics. The receiver performance shall be performance of the combination of the earphone element and receiver amplifier electronics. Receiver performance characteristics shall be measured under the conditions specified in 4.8.11. All receiver specifications shall apply for each polarity of transmitter dc current. The maximum difference in output level from positive to negative current operation shall be 1.5 dB.

3.6.1 Receiver output level. The 1000 Hz output level shall be 82 ±4 dB SPL when tested as specified in 4.8.11. When a headset with volume control is specified (see 3.1), volume control position 1 shall be used.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.

3.6.2 Receiver loudness level (RXLL). The receiver loudness level shall be 82 dB ±4 dB RXLL when tested as specified in 4.8.12. When a headset with volume control is specified (see 3.1), volume control position 1 shall be used.

3.6.3 Receiver volume control. When a headset with volume control is specified (see 3.1), the receiver volume control gain shall be as follows when tested as specified in 4.8.13:

a. Lowest level: Loudness level measurement of 3.6.2 or output level measurement of 3.6.1.

b. Mid level: Level +8 ±3 dB. (This measurement applies only to 3 position rocker switch controls.)

c. Highest level: Position 1 level +15 ±5 dB.

3.6.4 Compression level. When a headset with volume control is specified (see 3.1), the receiver compression level shall be 95 dB SPL ±5 dB when tested as specified in 4.8.14.

3.6.5 Receiver frequency response. When tested as specified in 4.8.15, the receiver frequency response of the headset shall fall within the envelope of figure 16.

3.6.6 Acoustic limiting. When tested as specified in 4.8.16, the frequency response shall fall on or below the envelope limits in figure 17.

3.6.7 Receiver linearity. When tested as specified in 4.8.17, the receiver output shall remain within the envelope limits of figure 18.

3.6.8 Common mode rejection. When tested as specified in 4.8.18, the receiver output level shall be at least 40 dB SPL below the level of paragraph 3.6.4.

3.6.9 Receiver distortion. When tested as specified in 4.8.19, the total harmonic distortion shall be 5 percent or less.

3.6.10 Transition times.

3.6.10.1 Automatic gain control (AGC) attack time. When tested as specified in 4.8.20.1, the AGC shall act to decrease the output to 120 percent of final level in 5 to 15 milliseconds.

3.6.10.2 Automatic gain control (AGC) release time. When tested as specified in 4.8.20.2, the AGC shall allow the output to rise to 80 percent of final level in 75 to 175 milliseconds.

3.6.10.3 Power-up time. The receiver output shall reach 80 percent of its final value within 90 milliseconds of the time that power is applied (see 4.8.20.3).

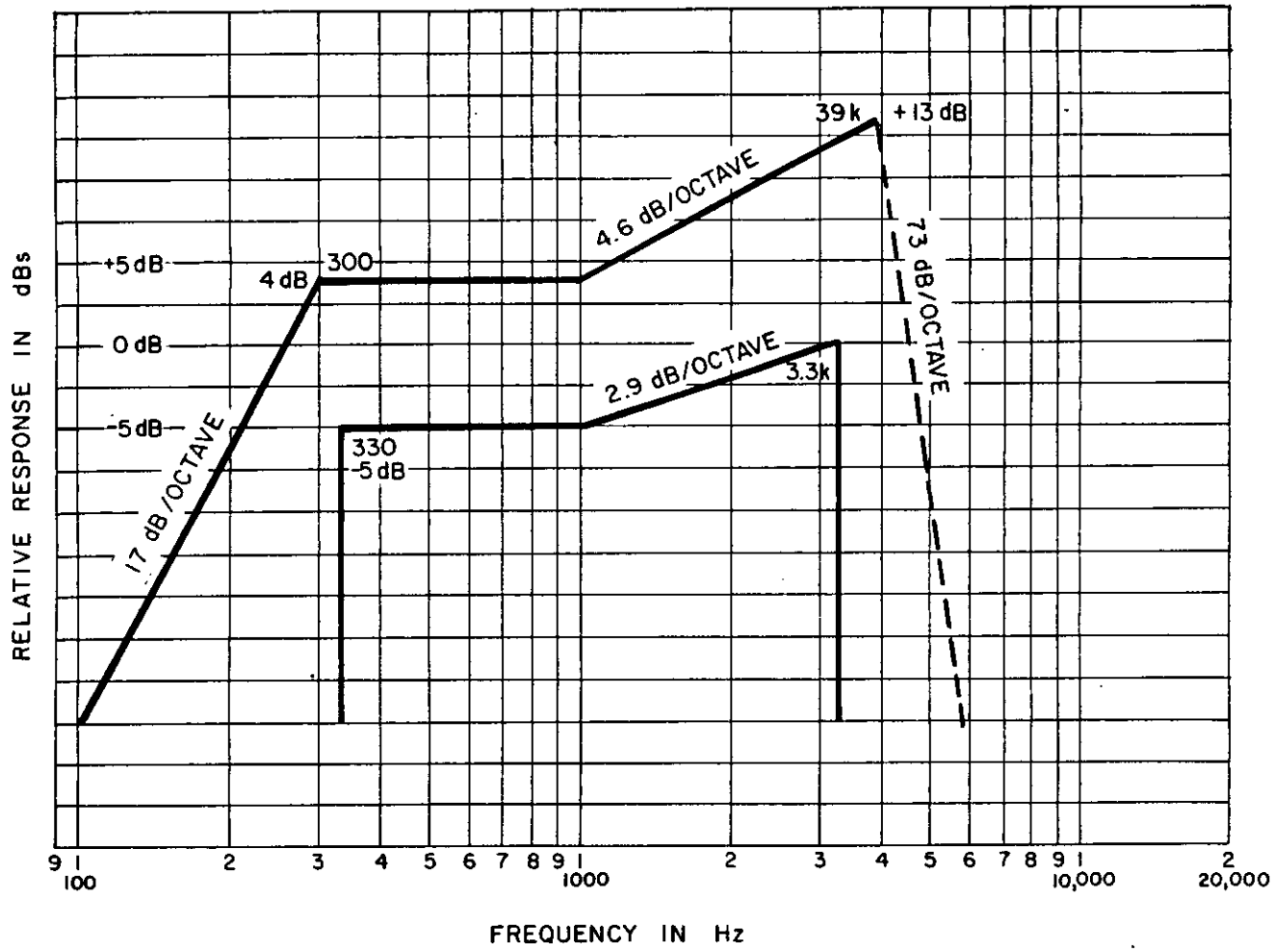
3.6.10.4 Power-down time (dual tone multiple frequency detection). When power is removed, the receive output shall remain within 6 dB of its original level for at least 100 milliseconds (see 4.8.20.4).

3.6.11 Input impedance. When tested as specified in 4.8.21, the receiver impedance shall be as specified (see 3.1).

3.7 Reverse sidetone. When tested as specified in 4.8.22, the transmitter output shall not exceed -46 dBV.

3.8 Insulation breakdown. The headset shall continue to meet specifications after applications of a 1000 V rms signal as specified in 4.8.23.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:45Z
Check the source to verify that this is the current version before use.



NOTE: The response curve shown is typical and may "float" within the enevelope.

FIGURE 16. Receive response envelope, intra-aural headset with and without volume control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

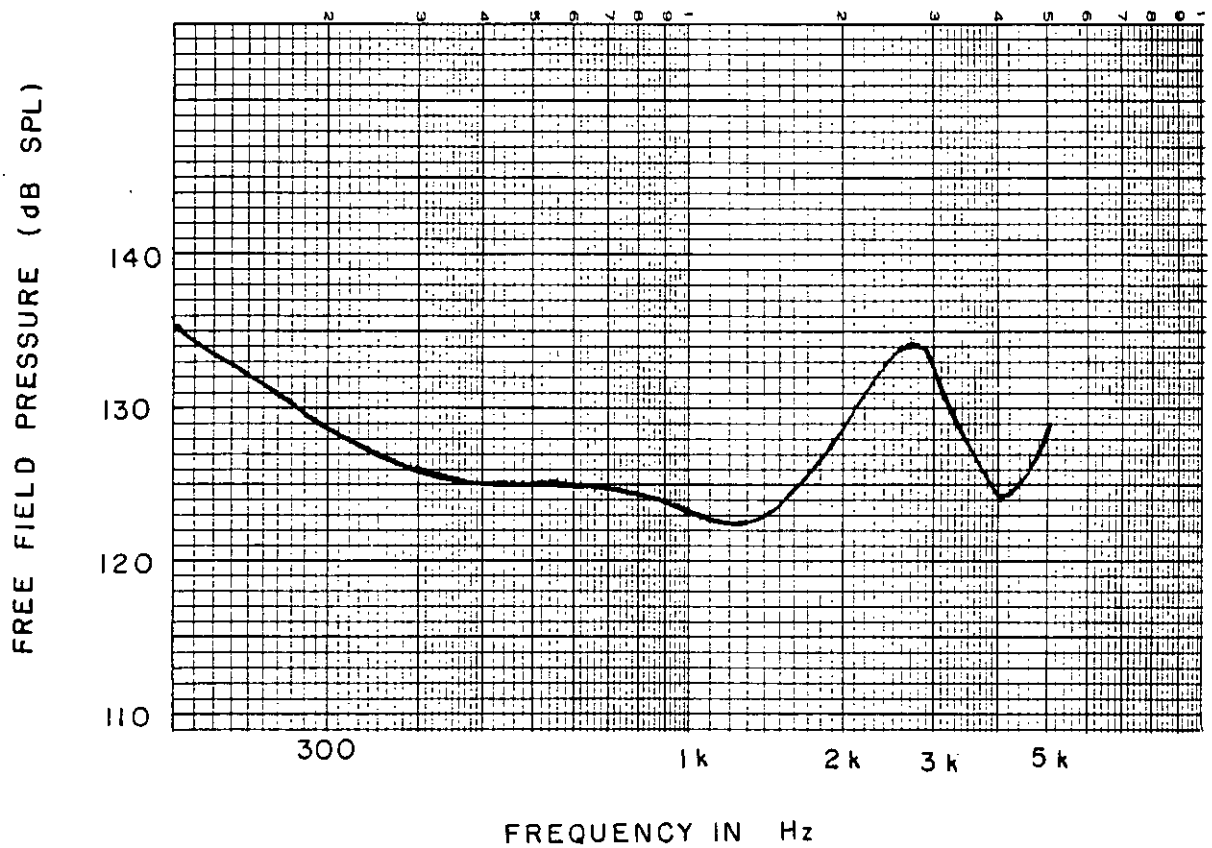


FIGURE 17. Acoustic limiting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

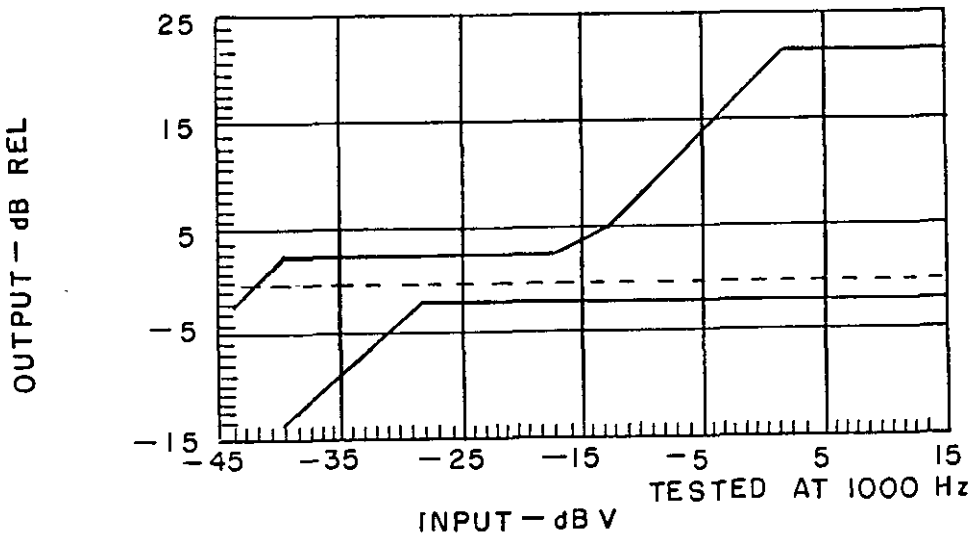


FIGURE 18. Receiver linearity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

3.9 Strain relief. When tested as specified in 4.8.24, there shall be no slippage of the cord or cable assembly out of the plug.

3.10 Static immunity. The headset shall continue to meet specifications after repeated static discharges when tested as specified in 4.8.25.

3.11 Environmental (see 4.8.26).

3.11.1 Temperature, humidity, and pressure. When tested in accordance with 4.8.26.1, the transmit and receive output levels shall not deviate by more than ±3 dB from their original levels and the response curves shall remain in their appropriate response envelopes.

3.11.2 Shock. When tested in accordance with 4.8.26.2, the transmit and receive output levels shall not deviate by more than ±3 dB from their original levels and the response curves shall remain in their appropriate response envelopes.

3.11.3 Vibration. When tested in accordance with 4.8.26.3, the transmit and receive output levels shall not deviate by more than ±3 dB from their original levels and the response curves shall remain in their appropriate response envelopes.

3.12 Cable flexing. When tested as specified in 4.8.27, the headset cable shall withstand 50,000 flexes while maintaining electrical integrity.

3.13 Marking. Headsets and headset-microphones shall be marked in accordance with MIL-STD-1285 with the Part or Identifying Number and manufacturer's code symbol or name.

3.14 Workmanship. Headsets shall be processed in such a manner as to be uniform in quality and shall be free from loose or deposited foreign materials, and other defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.3).

b. Qualification inspection (see 4.6).

c. Quality conformance inspection (see 4.7).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the equipment, are in accordance with the applicable referenced specifications prior to such fabrication.

TABLE I. Materials inspection.

Materials	Requirement paragraph	Applicable specification
Solder flux	3.3.1	MIL-F-14256

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Temperature: Room ambient, +19°C (+67°F) to +25°C (+77°F).

b. Pressure: Normal atmospheric.

c. Humidity: 45 to 50 percent relative humidity.

All acoustical testing shall be made in a free field environment.

4.5 Components inspection. Not applicable.

4.6 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.2) on sample units produced with equipment and procedures normally used in production.

4.6.1 Sample size. Six headset-microphones shall be subjected to the qualification inspection.

4.6.2 Inspection routine. The sample shall be subjected to the inspections specified in table II, in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided equally into two groups of three units each and subjected to the inspections for their particular group.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

4.6.3 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.6.4 Retention of qualification. To retain qualification, the contractor shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of the following:

a. A summary of the results of the tests performed for inspection of product for delivery, group A, indicating, as a minimum, the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

b. A summary of the results of tests performed for periodic inspection, group C, including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 12-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 6-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification. In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit the earphones to testing in accordance with the qualification inspection requirements.

4.7 Quality conformance inspection.

4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B.

4.7.1.1 Inspection lot. An inspection lot shall consist of all headset microphone assemblies of the same type produced under essentially the same conditions, and offered for inspection at one time.

4.7.1.2 Group A inspection. Group A inspection shall consist of the inspection specified in table III.

4.7.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be 1 percent major and 4 percent minor. Major and minor defects shall be as defined in MIL-STD-105.

4.7.1.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

TABLE II. Qualification inspection.

Inspection	Requirement paragraph	Test procedure paragraph
Group I		
Visual and mechanical	3.1, 3.3, 3.4, 3.13, and 3.14	4.8.1
Transmit performance		
Output level	3.5.1	4.8.2
Loudness level (TXLL)	3.5.2	4.8.3
Frequency response	3.5.3	4.8.4
Voice switching	3.5.4	4.8.5
Switch point	3.5.4.1	4.8.5.1
Switch depth	3.5.4.2	4.8.5.2
Warmup time	3.5.5	4.8.6
Distortion	3.5.6	4.8.7
DC resistance	3.5.7	4.8.8
Output impedance	3.5.8	4.8.9
Noise	3.5.9	4.8.10
Receiver performance		
Output level	3.6.1	4.8.11
Loudness level (RXLL)	3.6.2	4.8.12
Volume control	3.6.3	4.8.13
Compression level	3.6.4	4.8.14
Frequency response	3.6.5	4.8.15
Acoustic limiting	3.6.6	4.8.16
Linearity	3.6.7	4.8.17
Common mode rejection	3.6.8	4.8.18
Distortion	3.6.9	4.8.19
Transition time	3.6.10	4.8.20
AGC attack time	3.6.10.1	4.8.20.1
AGC release time	3.6.10.2	4.8.20.2
Power-up time	3.6.10.3	4.8.20.3
Power-down time	3.6.10.4	4.8.20.4
Input impedance	3.6.11	4.8.21
Group II		
Reverse sidetone	3.7	4.8.22
Insulation breakdown	3.8	4.8.23
Strain relief	3.9	4.8.24
Static immunity	3.10	4.8.25
Group III		
Environmental	3.11	4.8.26
Temperature, humidity, and pressure	3.11.1	4.8.26.1
Shock	3.11.2	4.8.26.2
Vibration	3.11.3	4.8.26.3
Cable flexing	3.12	4.8.27

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test procedure paragraph
Visual and mechanical	3.1, 3.3, 3.4, 3.13, and 3.14	4.8.1
Transmit performance		
Output level	3.5.1	4.8.2
Loudness level (TXLL)	3.5.2	4.8.3
Frequency response	3.5.3	4.8.4
Distortion	3.5.6	4.8.7
DC resistance	3.5.7	4.8.8
Output impedance	3.5.8	4.8.9
Receiver performance		
Output level	3.6.1	4.8.11
Loudness level (RXLL)	3.6.2	4.8.12
Volume control	3.6.3	4.8.13
Compression level	3.6.4	4.8.14
Frequency response	3.6.5	4.8.15
Distortion	3.6.9	4.8.19
Input impedance	3.6.11	4.8.21

4.7.1.3 Group B inspection. Group B inspection shall consist of the inspection specified in table IV and shall be made on sample units which have been subjected to and have passed the group A inspection.

4.7.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 2.5 percent defective.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test procedure paragraph
Transmit performance		
Voice switching	3.5.4	4.8.5
Switch point	3.5.4.1	4.8.5.1
Switch depth	3.5.4.2	4.8.5.2
Warmup time	3.5.5	4.8.6
Noise	3.5.9	4.8.10
Receiver performance		
Acoustic limiting	3.6.6	4.8.16
Linearity	3.6.7	4.8.17
Common mode rejection	3.6.8	4.8.18
Transition time	3.6.10	4.8.20
AGC attack time	3.6.10.1	4.8.20.1
AGC release time	3.6.10.2	4.8.20.2
Power-up time	3.6.10.3	4.8.20.3
Power-down time	3.6.10.4	4.8.20.4
Receiver sidetone	3.7	4.8.22

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.

4.7.1.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.7.1.3.3 Disposition of sample units. Sample units which have passed the group B inspection may be delivered on the contract or purchase order, if the lot is accepted and the sample units are still within specified electrical tolerances.

4.7.2 Periodic inspection. Periodic inspection shall consist of group C inspection. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.7.2.1.4), delivery of products which have passed groups A and B inspections shall not be delayed pending the results of these periodic inspections.

4.7.2.1 Group C inspection. Group C inspection shall consist of the inspections specified in table V in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspections.

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test procedure paragraph
Insulation breakdown	3.8	4.8.23
Strain relief	3.9	4.8.24
Static immunity	3.10	4.8.25
Environmental	3.11	4.8.26
Temperature, humidity, and pressure	3.11.1	4.8.26.1
Shock	3.11.2	4.8.26.2
Vibration	3.11.3	4.8.26.3
Cable flexing	3.12	4.8.27

4.7.2.1.1 Sampling plan. Four sample units shall be selected from each 2-months' production or each 1,000 units, whichever occurs first, after date of notification of qualification.

4.7.2.1.2 Failures. If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all the tests shall be repeated.

4.7.2.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

4.7.2.1.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured using essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.7.3 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

4.8 Methods of inspection.

4.8.1 Visual and mechanical inspection. Headsets shall be examined to verify that the materials, design, construction, physical dimensions, markings, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.13, and 3.14).

4.8.2 Transmit output level (see 3.5.1). The test circuit of figure 19 shall be used for measurement. Limits apply to test conditions 1 through 6 (see figure 19) at a frequency of 1000 Hz, unless stated otherwise. All transmit measurements shall be made using a sound field source having a frequency response flat within ±1 dB from 250 to 5000 Hz that can be maintained over a range of 64 to 114 dB SPL. Unless otherwise specified, the sound field shall be calibrated to 94 dB SPL, the normal test level, at 1/2 inch (12.7 mm) above the center of the mouth opening. The tip of the headset voice tube shall be placed at this distance. The tube shall be perpendicular to the mouth opening, and the tube and mouth opening shall be coaxial. The voice tube shall be fully retracted (see figure 20). The transmit output load shall be 50 ohms.

4.8.3 Transmit loudness level (TXLL) (see 3.5.2). Output level measurements shall be made at 11 discrete frequencies and shall be used to derive the transmit loudness level (TXLL). Measurements shall be made in accordance with 4.8.2. Test conditions are 1 and 2.

$$TXLL = 20 \log_{10} \left[\sum_{i=1}^{n-1} \left(\frac{V_i^{\frac{1}{2.2}} + V_{i+1}^{\frac{1}{2.2}}}{20} \right) \right]^{2.2}$$

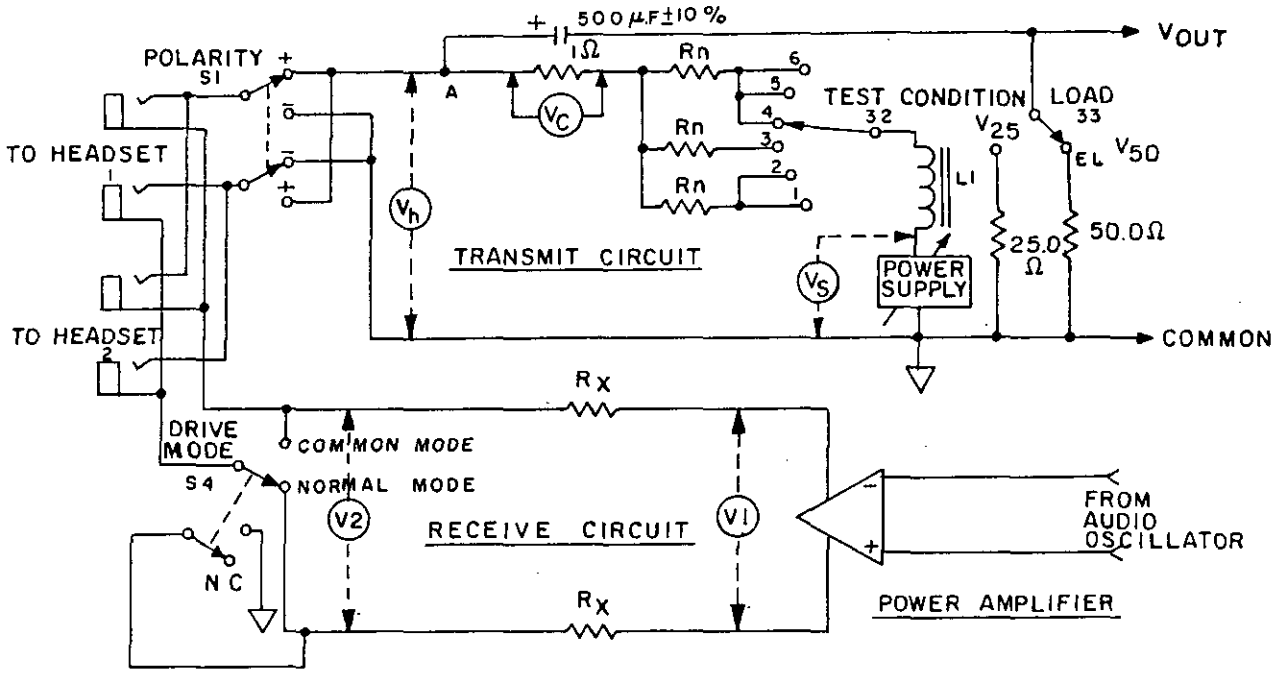
i	1	2	3	4	5	6	7	8	9	10	11
fi	315	400	500	630	800	1000	1250	1600	2000	2500	3150

i = Computational index.

f_i = Test frequency (Hz).

V_i = Output voltage (V rms).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.

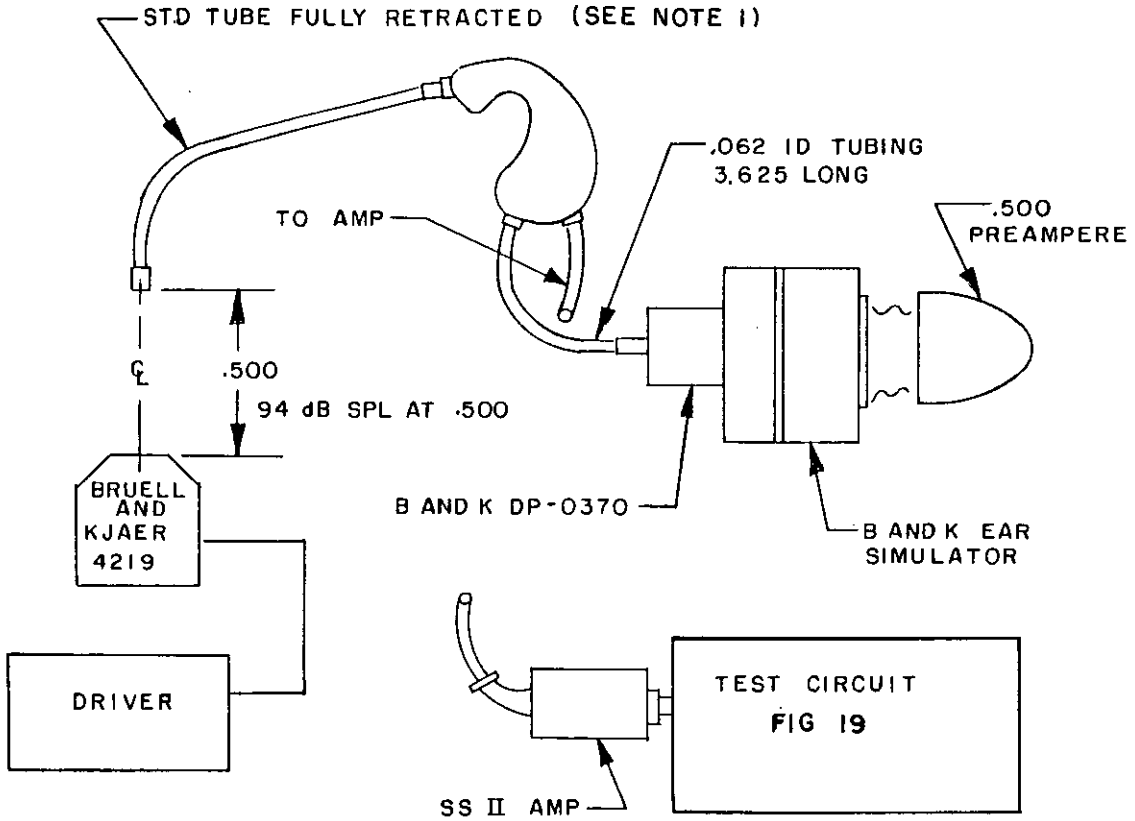


Test conditions	V_s ±5%	R_t ±1%	DCR HDST (9)	I_h (9)
1	4.8 V	226.5 ohms	120 ohms	14.0 mA
2	7.0 V	"	100 ohms	21.1 mA
3	4.0 V	315.5 ohms	400 ohms	5.5 mA
4	24.0 V	163.5 ohms	40 ohms	118.0 mA
5	20.0 V	"	45 ohms	96.0 mA
6	28.0 V	"	34 ohms	141.0 mA

NOTES:
1. All resistors ±1 percent tolerance.
2. R_t is the total dc resistance between A and B.
3. $R_n = R_t$ - 1 ohm - DCR (L1) - R (power supply).
4. L1 is greater than or equal to I_h, DCR is less than or equal to 60 ohms at 200 mA, Z is greater than or equal to 1000 ohms at 200 Hz.
5. Power supply: 3.0 to 30.0 V dc, current limited to 150 mA maximum.
6. Power amplifier: Output power is greater than or equal to 0.5 watt, output Z is less than or equal to 10 ohms.
7. Headset dc resistance may be calculated from: HDST DCR = V_h (in volts)/ I_h (in amperes).
8. V_C is numerically equal to the dc current through the headset (I_h).
9. I_h is a function of R_t, supply voltage, and the headset DCR; values of I_h and DCR listed are typical.
10. R_x = 150 ohms for 300-ohm receive impedance.
    R_x = 300 ohms for 600-ohm receive impedance.

FIGURE 19. Test circuit.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:45Z
Check the source to verify that this is the current version before use.



Inches	mm
.062	1.57
.500	12.70
3.625	92.08

NOTES:
1. Acoustic tube is coaxial with mouth opening.
2. Dimensions are in inches.
3. Metric equivalents are for general information.

FIGURE 20. Transmitter and receiver test system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

4.8.4 Frequency response (see 3.5.3). The 200 Hz to 5000 Hz response shall be measured as specified in 4.8.2 except the sound field shall be calibrated to 87 dB SPL.

4.8.5 Voice switching (see 3.5.4).

4.8.5.1 Switch depth (see 3.5.4.1). The test specified in 4.8.2 shall be performed except the test level shall vary from 64 to 94 dB SPL. The level at which switching occurs shall be recorded.

4.8.5.2 Switch depth (see 3.5.4.2). The test specified in 4.8.3.2 shall be performed. The transmit output level, switch-up time, and switch-down time shall be recorded.

4.8.6 Warmup time (see 3.5.5). The test specified in 4.8.3 shall be performed except the power applied to the headset shall be switched from off to on. The time required for the headset to reach 71 percent of the steady state value shall be recorded.

4.8.7 Distortion (see 3.5.6). Distortion measurements shall be made at 300, 1000, and 3000 Hz according to 4.8.2 except the acoustic input at 3000 Hz shall be 84 dB SPL.

4.8.8 DC resistance (see 3.5.7). The dc resistance in ohms shall be V_h (in volts)/I_h (in amperes) (see figure 21).

4.8.9 Output impedance (see 3.5.8). The output impedance (Zout) in ohms shall be:

$$\frac{50(V_{50} - V_{25})}{(2V_{25} - V_{50})}$$ (see figure 19).

4.8.10 Transmit noise (see 3.5.9). Transmit noise shall be measured across the output load (test condition 4) using a Wilcom T-194C or equivalent transmission and noise test set. Connect output to line I T and R terminals of set. Switches shall be set as follows:

a. Line select: RCV I/SEND II.

b. Impedance: 900.

c. Term/bridge: BRDG.

d. Meter range: -3 to +3.

e. Meter response: NORMAL.

f. Weighting: C-MSG.

4.8.11 Receiver output level (see 3.6.1). All receiver parameters shall be measured with a Bruel and Kjaer 4157 ear simulator fitted with a DP-0370 earplug simulator (see figure 20). The headset under test is attached to the coupler with a standard eartube (.062 inch inside diameter) (see figure 21). Unless stated otherwise, the headset receiver drive level at V1 shall be -38 dBV and the test frequency shall be 1000 Hz. Unless otherwise specified, all specifications shall apply to test condition 3 at the lowest volume control position. The measurement configuration of 4.8.2 shall apply.

4.8.12 Receiver loudness level (RXLL) (see 3.6.2). Output level measurements shall be made at 11 discrete frequencies and used to derive the receiver loudness level (RXLL). Measurements shall be made in accordance with 4.8.11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

$$RXLL = 20\ LOG_{10}\left[\sum_{i=1}^{n-1}\left(\frac{10^{\frac{P_i}{44}} + 10^{\frac{P_{i+1}}{44}}}{20}\right)\right]^{2.2}$$

i	1	2	3	4	5	6	7	8	9	10	11
f_i	315	400	500	630	800	1000	1250	1600	2000	2500	3250
X_i	0	-.9	-1.2	-1.2	-1.2	-1.3	-1.5	-2.2	-3.6	-6.3	-10.8

i = Computational index.

f_i = Test frequency.

X_i = In-ear canal correction factor (dB).

Pi = Effective acoustical output (dB). Pi is calculated by measuring the sound pressure output (in dB P_a) at Fi and adding the correction factor X_i.

4.8.13 Receiver volume control (see 3.6.3). The test specified in 4.8.11 or 4.8.12 shall be performed. The lowest and highest volume control positions shall be utilized. The dB differences in the output levels shall be recorded.

4.8.14 Compression level (see 3.6.4). The test specified in 4.8.4 shall be performed except the drive level shall be -15 dBV.

4.8.15 Receiver frequency response (see 3.6.5). The test specified in 4.8.4 shall be performed except the drive shall be swept from 100 to 6000 Hz.

4.8.16 Acoustic limiting (see 3.6.6). The test specified in 4.8.4 shall be performed except the drive level shall be +15 dBV.

4.8.17 Receiver linearity (see 3.6.7). The test specified in 4.8.11 shall be performed except the 1000 Hz drive level shall vary from -45 dBV to +15 dBV.

4.8.18 Common mode rejection (see 3.6.8). A -4 dBV, 1000 Hz drive shall be applied across the simplexed headset sleeves. The receiver output shall be measured according to 4.8.11.

4.8.19 Receiver distortion (see 3.6.9). The test specified in 4.8.11 shall be performed except the test frequencies shall be 300, 1000, and 3000 Hz and the receiver volume control shall be in the lowest position.

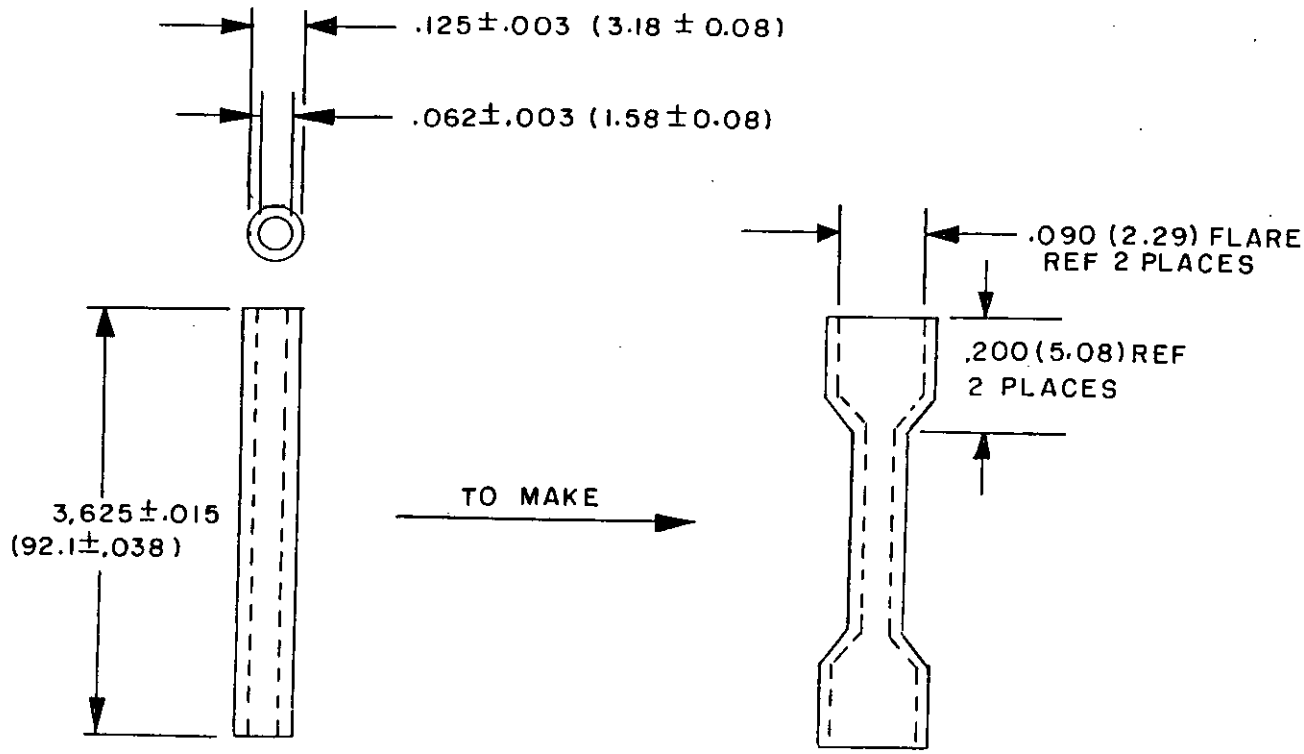
4.8.20 Transition time.

4.8.20.1 AGC attack time (see 3.6.10.1). The test specified in 4.8.11 shall be performed except that the drive level shall be step increased from a level below the compression level to a level above the threshold.

4.8.20.2 AGC release time (see 3.6.10.2). The test specified in 4.8.11 shall be performed except the drive level shall be step decreased from a level above the threshold to a level below the threshold.

4.8.20.3 Power-up time (see 3.6.10.3). The test specified in 4.8.11 shall be performed except that the supply power shall be turned from off to on. The time required for power-up shall be recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Material: Vinyl formulation B-44-3 food and beverage tygon tubing (PLX. P/N 07173-00).
4. Vendor: Norton Plastics and Synthetics Division, Akron, Ohio, or equal.

FIGURE 21. Standard ear tube assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

4.8.20.4 Power-down time (see 3.6.10.4). The test specified in 4.8.11 shall be performed except that the supply power shall be turned from on to off. The time required for power-down shall be recorded.

4.8.21 Input impedance (see 3.6.11). Receiver impedance shall be calculated from Z = 300 V2/(V1-V2) for 4-wire headset and 600 V2/(V1-V2) for 6-wire headset from 300 to 3000 Hz (see figure 21).

4.8.22 Reverse sidetone (see 3.7). With the microphone acoustic tube sealed, and a receiver drive of -12 dBV, the transmit output level shall be tested according to 4.8.2 except the frequencies shall be from 200 to 10,000 Hz.

4.8.23 Insulation breakdown (see 3.8). Five 10-second applications of a 1000 V rms, 60 Hz signal shall be applied between the simplexed tip/sleeve leads and all exposed metal parts.

4.8.24 Strain relief (see 3.9). Headset cables shall receive a minimum of 10 pounds pull for 1 second.

4.8.25 Static immunity (see 3.10). Repeated static discharges shall be applied to the outer surfaces of the headset under simulated normal conditions. The types of discharge shall include: 50 discharges of each polarity, through a 10,000 ohms resistor connected to a 60 picofarads capacitor charged to 20 kilovolts; and 20 discharges of each polarity, through a 500 ohms resistor connected to a 100 picofarads capacitor charged to 10 kilovolts.

4.8.26 Environmental (see 3.11).

4.8.26.1 Temperature, humidity, and pressure (see 3.11.1). The headset shall be exposed to the following conditions:

a. Storage temperature, humidity, and pressure. The headset shall be returned to standard environmental conditions from the storage conditions, after a suitable drying out period. Dwell time at each extreme shall be 48 hours.

(1) Temperature range: -40°C to +65°C.

(2) Humidity range: 5 to 95 percent RH.

(3) Pressure range: 8.7 to 14.7 psia.

b. Operating temperature, humidity, and pressure. The headset shall meet the requirements of 3.11.1 under the following environmental conditions:

(1) Temperature range: 0°C to +50°C.

(2) Humidity range: 5 to 95 percent RH (noncondensing).

(3) Pressure range: 9.4 to 14.7 psia.

c. Temperature and humidity cycling (storage). The headset shall meet the requirements of 3.11.1 after being cycled at any convenient rate three times through the following conditions:

(1) 30 minutes at 150°F and 15 percent RH.

(2) 30 minutes at 90°F and 90 percent RH.

(3) 30 minutes at -40°F and any convenient RH.

Condensation shall be avoided and the headset dried out before operation.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the currentversion before use.

4.8.26.2 Shock (see 3.11.2). The headset shall be subjected to 18 random drops onto an asphalt tile-covered concrete floor. Drop height for the capsule shall be 54 inches; for the amplifier, 30 inches.

4.8.26.3 Vibration (see 3.11.3). The headset shall be exposed to 5g peak (displacement less than .03 inch) sinusoidal vibration along each of three orthogonal axes. Frequency shall be swept logarithmically at 0.1 octave per minute from 5 to 100 Hz and 0.25 octave per minute from 100 to 500 Hz for a total of approximately 1 hour in each axis.

4.8.27 Cable flexing (see 3.12). The cable shall be run through 50,000 flex cycles over a 1/4-inch radius, 60° to each side of vertical at a rate of 85 flex cycles per minute with a load of 12 ±1/4 ounce, attached to the lower end of the cable. One cycle equals two complete 120° arcs of travel. The 100,000 flex cycles shall be computed as arithmetic mean of five test samples, the lowest single sample life being 66,000 flex cycles. Failure of the cable shall be considered as a 25 percent increase in the initial resistance of the test sample in which the four or six conductors are connected in series.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C, as specified (see 6.3).

5.1.1 Level A.

5.1.1.1 Cleaning. Headset-microphone assemblies shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Headset-microphone assemblies shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packs. Each headset-microphone assembly shall be individually unit packed in accordance with submethod IA-8 of MIL-P-116 insuring compliance with the applicable requirements of that specification. Cushioning shall be noncorrosive and shall not crumble, flake, powder, or shed. Each unit pack shall be placed in a supplementary container conforming to PPP-B-636, class weather resistant.

5.1.1.5 Intermediate packs. Intermediate packs are not required.

5.1.2 Level B. The requirements for level B shall be as specified for level A except that submethod IC-1 or IC-3 of MIL-P-116 shall be substituted for submethod IA-8 and the supplementary container shall conform to PPP-B-636, class domestic.

5.1.3 Level C. The level C preservation of headset-microphone assemblies shall conform to the MIL-STD-2073-1 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.3).

5.2.1 Level A. Headset-microphone assemblies, preserved as specified in 5.1, shall be packed in wood boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Closure and strapping shall be in accordance with the applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish A. The requirements for level B packing shall be used when the total quantity of a stock numbered assembly for a single destination does not exceed a packed volume of 1 cubic foot (0.0283 cubic meter).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Headset-microphone assemblies, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. The requirements for box closure, waterproofing, and reinforcing shall be in accordance with method V of the PPP-B-636 appendix.

5.2.3 Level C. Headset-microphone assemblies, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet (1.1328 cubic meters) or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Headset-microphone assemblies, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a wood cap (storage aid 5) positioned over each load.

5.2.4.2 Level B. Headset-microphone assemblies, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that weather resistant fiberboard caps (storage aid 4) shall be used in lieu of wood caps.

5.2.4.3 Level C. Headset-microphone assemblies, packed as specified in 5.2.3, shall be unitized as specified in 5.2.4.2 except that the fiberboard caps shall be class domestic.

5.3 Marking. In addition to any special or other identification marking required by the contract (see 6.3), each unit, supplementary, and exterior container, and unitized load shall be marked in accordance with MIL-STD-129. The complete military or contractor's type or part number, as applicable (including the CAGE), shall be marked on each unit and supplementary pack in accordance with the identification marking provisions of MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.7.3.

6. NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Headset-microphone, as specified herein, are intended for use in applications, such as, air traffic control, telecommunications operators, and video display terminal operations where the ambient noise level is below 85 dB SPL.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Levels of preservation and packing required (see 5.1 and 5.2).

c. If special or additional identification marking is required (see 5.3).

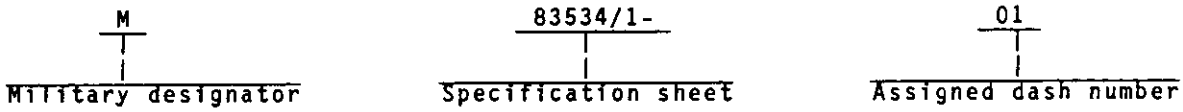
6.3 Packaging requirements. The preservation, packing, and marking herein are intended for direct shipments to the Government. However, at the option of the contractor, or when so specified, the packaging provisions herein are also applicable for the preparation of headset-microphone assemblies for shipment from the parts contractor to the original equipment manufacturer.

6.4 Condition for use of level B preservation. When level B preservation is specified (see 5.1.2), this degree of protection should be used for the acquisition of headset-microphone assemblies for resupply worldwide under known favorable handling, transportation, and storage conditions.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contracts, qualified for inclusion in applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the qualified products list is the Air Force 2750th ABW, ATTN: ESP Gentile AFS, OH 45444; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), Dayton, OH 45444. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6". Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.6 Anechoic room or chamber (free field room or environmental). An anechoic or free field environment is a room or area whose boundaries absorb effectively all the sound incident thereon, thereby affording essentially free field conditions.

6.7 Part or identifying number (PIN). The pin shall consist of the letter "M" the basic number of the applicable specification sheet, and an assigned dash number as shown in the following example (see 3.1 and 6.2):



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

6.8 Subject term (key word) listing.

Ear plug
Headset, lightweight
Intraaural
Voice tube

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5965-F194)

☆U.S. GOVERNMENT PRINTING OFFICE: 1989 - 704-034/14242

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

2750 ABW/ES
GENTILE AF STATION
DAYTON, OH 45444-4500

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:45Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:45Z
Check the source to verify that this is the current version before use.